

Deep Reinforcement Learning for Traffic Light Control

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Abstract—This document is a model and instructions for L^AT_EX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.

Index Terms—component, formatting, style, styling, insert.

I. INTRODUCTION

- What are SUMO limitations? - Do we optimize traffic lights only for cars or also bikes, pedestrians, etc. - Do we focus on multiple intersections or just one? - Are we using acyclical or cyclical light switches? May one of them be more realistic than the other? - Do we train on multiple scenarios, like “rush hours” and low traffic conditions, or do we only have normal traffic? - Do we use DQN or are there also other choices like PPO (is PPO able to combine with neural nets?)?

a good starting tutorial on how to generate traffic scenarios can be found here [1].

- at the end, compare normal (static) traffic light with our solution. if possible, the static traffic light control should be decently good. maybe its possible to use a real example?

II. NETWORKS

A. Jönköping

map of sweden downloaded here:
<https://download.geofabrik.de/europe/sweden.html> geojson
polygon object created here: <https://geojson.io/>

then, we used osmium and osmfilter to apply the polygon shape to the map of sweden. afterwards, we filter only for highways, except for those of the following types:

- footway
- path
- pedestrian
- steps
- cycleway
- bridleway

finally, we use the SUMO *netconvert* tool to convert the map into a SUMO network.

1) *Traffic Assignment Zones*: Traffic Analysis Zones (TAZs) are spatial units used in transportation modeling and traffic studies to represent areas with similar travel behavior. A TAZ typically models a neighborhood or district, and serves as a fundamental unit for estimating travel demand. Each zone can be associated with socio-economic and demographic



Fig. 1. The filter (in grey) that defines the boundaries of the road network in Jönköping.

data, such as population, employment, and land use. (COPIED FROM [1]).

since we have no information about the actual TAZ's in Jönköping, we create these zones randomly with the *gridDistricts* tool provided by SUMO.

2) *Rush Hours and Noise*:

REFERENCES

- [1] D. A. Guastella, E. Montero-Porras, A. Morales-Hernández, and G. Bontempi, “Traffic modeling with sumo: a tutorial,” 2025. [Online]. Available: <https://arxiv.org/abs/2304.05982>